01. Foundations of Parallel Computing & Performance Measures

This chapter introduces the fundamental concepts, performance metrics, and basic patterns (skeletons) used in parallel computing to evaluate and structure concurrent execution.

1. Performance Metrics

To evaluate the quality and efficiency of a parallel system, we rely on several quantitative metrics:

1.1 Latency (L)

Latency is the time required to complete a single task from its beginning to its end. In a system composed of multiple stages (e.g., a pipeline), it is the sum of the times spent in each individual stage.

1.2 Service Time (T_s)

Service Time is the time interval between the emission of two consecutive results. It dictates the maximum throughput of the system. * In a sequential execution, T_s is equal to the Latency. * In a parallel system (like a pipeline), the overall service time is bottlenecked by the slowest stage: $T_s = \max\{T_{s_1}, T_{s_2}, \ldots, T_{s_k}\}$.

1.3 Completion Time (T_c)

The total time required to process m tasks. Given a pipeline or a generic parallel system, the initial phase involves filling the system (paying the latency L), and subsequently, results are emitted every T_s.

$$T_c = L + (m-1)T_s$$

For a very large number of tasks ($m \gg 1$), the Completion Time can be approximated as $T_c \approx mT_s$.

1.4 Speedup (S_p)

Speedup measures the relative performance gain of executing a program on n parallel processing elements compared to a sequential execution.

$$S_p(n) = \frac{T_{seq}}{T_{par}(n)}$$

Where T_{seq} is the completion time of the best sequential algorithm and $T_{par}(n)$ is the completion time of the parallel algorithm using n workers. * Ideally, we seek **linear speedup** ($S_p(n) = n$). * Speedup is theoretically bounded by the critical path: $S_p \leq \frac{\text{Work}}{\text{Span}}$.

1.5 Efficiency (E)

Efficiency measures how well the parallel resources are being utilized. It is defined as the ideal parallel time divided by the actual parallel time, or equivalently, Speedup divided by the number of workers n.

$$E(n) = \frac{T_{id}(n)}{T(n)} = \frac{T_{seq}}{n \cdot T_{par}(n)} = \frac{S_p(n)}{n}$$

An ideal efficiency is 1 (or 100%), corresponding to linear speedup.

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2. Laws of Scalability

2.1 Amdahl's Law

Amdahl's law provides a theoretical upper bound on the speedup of a program when only a fraction of it can be parallelized. If f is the strictly sequential fraction of the program (e.g., $10\% = 0.10$), the maximum

speedup is bounded by $\frac{1}{f}$ (e.g., 10×) regardless of how many cores n are added.

$$S_p(n) \leq \frac{1}{f + \frac{1-f}{n}} \xrightarrow{n \to \infty} \frac{1}{f}$$

2.2 Gustafson's Law

Gustafson's Law counteracts the pessimism of Amdahl's Law by observing that as computing power increases, the size of the problem typically increases as well. The work required for the parallel part grows much faster than the sequential part. If the sequential fraction f decreases as the data size grows, the system can achieve near-linear speedup for massive datasets.

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3. Basic Parallel Patterns

Parallel patterns (or algorithmic skeletons) provide high-level abstractions to structure concurrent computations, avoiding low-level lock and thread management.

3.1 Pipeline

A computation is divided into k sequential stages connected by communication channels. Each stage processes a data item and passes it to the next. * **Service Time (T_s):** Bounded by the slowest stage. $T_s = \max_{i=1\ldots k}\{T_{s_i}\}$. * **Latency ($L$):** Sum of the times of all stages. $L = \sum_{i=1}^{k} T_{s_i}$. * **Optimization:** If a stage is a bottleneck, its service time can be lowered by parallelizing that specific stage internally (e.g., turning it into a Farm).

3.2 Farm (Task Farm / Master-Worker)

A Farm replicates the execution of a single function over multiple workers. It is ideal for embarrassingly parallel workloads where tasks are completely independent. * **Structure:** Consists of an *Emitter* (Scheduler) that distributes tasks, a set of *Workers* that compute the tasks independently, and a *Collector* that gathers the results. * **Service Time (T_s):** If the sequential service time is T_w and we have n_w workers, the ideal parallel service time is $T_s = \frac{T_w}{n_w}$. * **Efficiency:** A farm is highly efficient, but scheduling overhead and the sequential nature of Emitter/Collector can become bottlenecks if the communication time dominates the computation time.

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4. Hardware & Context

When designing parallel applications, the underlying hardware heavily influences the chosen mechanism and abstraction: * **Memory Architectures:** Threads can communicate via Shared Memory (requiring synchronization primitives like locks, condition variables, or barriers) or via Message Passing (Point-to-Point or Collective communications). * **Grain Size (Granularity):** The ratio of computation time to communication time. A coarse-grained computation (high computation-to-communication ratio) is generally necessary to offset the overhead of data splitting and dispatching. * **Excess Parallelism (Oversubscription):** Spawning more parallel activities than the available hardware contexts. It allows the OS/runtime to hide latency (e.g., if one thread blocks for I/O, another can be scheduled), avoiding idle cores.

02. Structured Parallel Programming & Algorithmic Skeletons

This chapter delves into Structured Parallel Programming (SPP) and Algorithmic Skeletons. The core philosophy of this approach is the **separation of concerns**: isolating the purely sequential computational logic (business logic) from the parallel coordination and communication mechanisms.

1. The Algorithmic Skeleton Framework

Introduced by Murray Cole, **Algorithmic Skeletons** are recurring parallel paradigms abstracted into reusable, parameterizable templates. They provide high-level parallel constructs that hide the underlying complexity of thread management, synchronization, and message passing.

Skeletons are typically divided into two macro-categories based on the nature of the parallelism they exploit:

1. **Stream Parallel Skeletons:** Operate on streams of data items (tasks) flowing through the system. Examples: `pipeline`, `farm`.
2. **Data Parallel Skeletons:** Operate on large, partitioned data structures where the same computation is applied to different chunks simultaneously. Examples: `map`, `reduce`, `stencil`.

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2. Stream Parallel Skeletons

Stream parallel skeletons model systems that process a continuous flow of input tasks. The execution is driven by the availability of new data tokens.

2.1 The Pipeline Skeleton (`pipe`)

The `pipe` skeleton connects k stages $S_1, S_2, \ldots, S_k$ in a sequence. The output stream of stage S_i becomes the input stream of stage S_{i+1}.

- **Semantics:** $\text{pipe}(S_1, S_2)(x) = S_2(S_1(x))$ extended over a stream.
- **Performance Model:**
  - **Service Time (T_s):** Bottlenecked by the slowest stage. $T_s = \max_{i=1\ldots k}\{T_{S_i}\}$
  - **Latency (L):** The sum of the times taken by all individual stages. $L = \sum_{i=1}^{k} T_{S_i}$
- **Optimization:** If a stage is identified as a bottleneck, it must be parallelized (e.g., by wrapping it in a `farm` or a nested `pipe`) to lower its service time and improve the overall throughput.

2.2 The Farm Skeleton (`farm`)

The `farm` skeleton models independent task execution (embarrassingly parallel). It consists of three entities:

1. **Emitter (E):** Fetches data from the input stream and dispatches it to the workers.
2. **Workers ($W_1 \ldots W_n$):** A pool of threads/processes computing the identical function f on different input data.
3. **Collector (C):** Gathers results from the workers and forwards them to the output stream.

- **Performance Model:**
  - Let T_E be the time the Emitter takes to schedule a task, T_W the time a worker takes to compute f, and T_C the time the Collector takes to gather a result.
  - **Service Time (T_s):** $T_s = \max\left\{T_E, \frac{T_W}{n}, T_C\right\}$
  - **Latency (L):** $L = T_E + T_W + T_C$
- **Scalability Bottleneck:** The Emitter and Collector are strictly sequential components. As the number of workers n increases, $\frac{T_W}{n}$ decreases. Eventually, T_E or T_C will become the maximum value, causing the speedup to plateau. The maximum useful number of workers is $n_{max} = \frac{T_W}{\max(T_E, T_C)}$.

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3. Data Parallel Skeletons

Data parallel skeletons apply operations across partitions of a single, large data structure (like vectors, matrices, or graphs).

3.1 The Map Skeleton (`map`)

The `map` skeleton takes an array (or list) and applies a purely sequential function f to each element independently.

- **Semantics:** $\text{map}(f)([x_1, x_2, \ldots, x_n]) = [f(x_1), f(x_2), \ldots, f(x_n)]$
- **Execution:** The runtime partitions the array into chunks, assigns chunks to different processing elements (PEs), computes the chunks in parallel, and merges the results.
- **Completion Time (T_c):** For an array of size N distributed over n workers, the ideal completion time is $T_c = \frac{N \cdot T_f}{n} + T_{scatter} + T_{gather}$.

3.2 The Reduce Skeleton (`reduce`)

The `reduce` skeleton takes an array and aggregates its elements into a single value using a binary operator $\oplus$. For the skeleton to yield deterministic results regardless of the parallel execution tree, the operator $\oplus$ **must be associative**.

- **Semantics:** $\text{reduce}(\oplus)([x_1, x_2, \ldots, x_n]) = x_1 \oplus x_2 \oplus \cdots \oplus x_n$
- **Execution:** Typically implemented using a parallel tree reduction.
- **Completion Time (T_c):** With n workers, the reduction tree has a depth of $\log_2(n)$. Thus, parallel time scales logarithmically rather than linearly in the number of processors.

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4. Operational Semantics for Skeletons

To formalize the behavior of these skeletons without ambiguity, we use operational semantics modeled as transition systems. States represent the current configuration of the computation, and transitions ($\rightarrow$) describe computation steps.

Let $\langle x, \tau \rangle$ represent an input stream where x is the current token and τ is the rest of the stream. Let Δ represent the inner computation of a skeleton (e.g., the worker function in a farm).

1. **Sequential execution (`seq`):**

   Applies a function f to a stream.

$$\langle \text{seq } f, \langle x, \tau \rangle \rangle \rightarrow \langle f(x) \rangle :: \langle \text{seq } f, \langle \tau \rangle \rangle$$

2. **Farm execution (`farm`):**

   The farm processes the first element of the stream and concurrently processes the rest.

$$\langle \text{farm } \Delta, \langle x, \tau \rangle \rangle \rightarrow \langle \Delta(x) \rangle \parallel \langle \text{farm } \Delta, \langle \tau \rangle \rangle$$

   (Note: The exact formalization can include explicit rules for Emitter scheduling and Collector gathering depending on the specific transition system used).

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5. Implementation in Real Frameworks (e.g., C++ FastFlow)

In modern C++ environments, implementing these mathematical models effectively requires low-overhead runtime supports. Frameworks like FastFlow implement stream parallel skeletons using **lock-free** and **wait-free** queues (e.g., Single-Producer Single-Consumer queues for pipelines).

- In a FastFlow `ff_pipeline`, each stage is a C++ object executing a sequential loop, pulling data from a lock-free input queue and pushing to an output queue.
- In a `ff_farm`, the Emitter acts as a load balancer, distributing pointers to memory allocations rather than copying the data itself, thereby minimizing T_E and extending the scalability limits defined by Amdahl's Law.

03. Shared Memory & Low-Level Concurrency

This chapter explores the intricacies of Shared Memory architectures, the challenges of keeping memory consistent across multiple processors, and the practical implementation of concurrent execution using modern tools.

1. Shared Memory Architectures

In a shared memory system, multiple processing elements (PEs) have access to a single, global memory address space. While this simplifies the programming model (data does not need to be explicitly sent and received via messages), it introduces severe hardware and software complexities.

1.1 Cache Coherence

Modern CPUs utilize multiple levels of cache (L1, L2, L3) to hide main memory latency. If multiple cores load the same memory block into their local L1 caches and one core modifies it, the other copies become stale.

- **Coherence Protocols:** The hardware must keep all these copies consistent and coherent through specific protocols (e.g., MESI - Modified, Exclusive, Shared, Invalid).
- **Performance Impact:** When a core writes to a shared variable, the coherence protocol must invalidate or update the cached copies in other cores. This generates heavy bus traffic and latency, a phenomenon known as **False Sharing** if cores are writing to different variables that happen to reside on the same cache line.

1.2 Synchronization Primitives

To prevent race conditions when multiple threads access shared data, we must enforce mutual exclusion using synchronization mechanisms:

- **Mutexes and Locks:** Ensure that only one thread can execute a critical section at a time. However, locking forces sequential execution, impacting Amdahl's Law and adding overhead.
- **Atomics and Lock-Free Programming:** Utilizing hardware-level atomic instructions (like Compare-And-Swap) to update variables without blocking threads. This is crucial for high-performance concurrent data structures.

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2. Branch Divergence and Predication

When executing parallel workloads on architectures with SIMT (Single Instruction, Multiple Threads) execution models, such as GPUs or highly vectorized CPU units, control flow becomes a critical bottleneck.

2.1 The Issue with Branching

If threads executing in lockstep encounter a conditional branch (e.g., an `if-else` statement) and diverge (some take the `true` path, others the `false` path), the hardware must serialize the execution. It first executes the `true` path for the active threads, masking the others, and then executes the `false` path. This halves the parallel efficiency.

2.2 Predication as a Solution

To mitigate branch divergence, one technique is **predication**. Instead of branching, the hardware computes both paths simultaneously. * **How it works:** All threads compute the instructions for both the `if` and the `else` branches. The results are guarded by a boolean predicate (flag). Only the results associated with the `true` predicate are actually committed to memory or registers. * **Trade-off:** This requires doing extra computational work, but it avoids the severe penalty of breaking the instruction pipeline and serializing the parallel execution context.

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3. Concurrency in Modern C++

Modern standards provide robust abstractions for managing shared memory parallelism without relying on raw OS-level threads (like POSIX threads).

3.1 Standard Threads and Data Races

Managing threads directly requires explicit synchronization.

```cpp
#include <iostream>
#include <thread>
#include <vector>
#include <mutex>

std::mutex mtx;

void compute_chunk(int thread_id, int& shared_accumulator) {
    // compute local aggregate to minimize lock contention
    int local_sum = 0;
    for (int i = 0; i < 1000; ++i) {
        local_sum += (i * thread_id);
    }

    // update shared state safely
    std::lock_guard<std::mutex> lock(mtx);
    shared_accumulator += local_sum;

    std::cout << "THREAD " << thread_id << " FINISHED EXECUTION. CURRENT ACCUMULATOR: " << shared_accum
}

int main() {
    int shared_accumulator = 0;
    std::vector<std::thread> pool;

    // spawn threads
    for (int i = 0; i < 4; ++i) {
        pool.emplace_back(compute_chunk, i, std::ref(shared_accumulator));
    }
```

```
    // wait for completion
    for (auto& t : pool) {
        t.join();
    }

    std::cout << "FINAL RESULT COMPUTED SUCCESSFULLY\n";
    return 0;
}
```

3.2 OpenMP

For data-parallel loops and regular computational kernels, OpenMP provides compiler directives (`#pragma`) that dramatically simplify parallelization by handling thread creation, work distribution, and synchronization automatically.

```
#include <iostream>
#include <vector>
#include <omp.h>

void process_array(std::vector<int>& data) {
    int total_sum = 0;

    // parallelize the loop and perform a reduction on total_sum
    #pragma omp parallel for reduction(+:total_sum)
    for (size_t i = 0; i < data.size(); ++i) {
        data[i] = data[i] * 2;
        total_sum += data[i];
    }

    std::cout << "OPENMP REDUCTION COMPLETED. SUM: " << total_sum << "\n";
}
```

By leveraging these constructs, developers can bridge the gap between abstract algorithmic skeletons (like Map and Reduce) and actual high-performance execution on shared memory architectures.

04. Thread Affinity & Memory Consistency

This chapter explores the low-level interactions between the operating system, the hardware architecture, and concurrent programs. Understanding how threads are scheduled and how memory is accessed is crucial for extracting maximum performance from multi-core systems.

1. Thread and Process Affinity

When the operating system's scheduler manages a pool of threads, it is free to migrate them across different physical cores to balance the system load. While this is optimal for general-purpose computing, it introduces severe performance penalties for High-Performance Computing (HPC) and structured parallel applications.

1.1 The Cost of Thread Migration

If a thread is executing on Core C_0 and is migrated to Core C_1, the following issues arise:

- **Cache Invalidation:** The L1 and L2 caches of C_0 contain the hot data for the thread. When moved to C_1, the thread experiences a burst of cache misses, forcing the CPU to fetch data from the slower L3 cache or Main Memory.

- **Context Switch Overhead:** The registers, program counter, and translation lookaside buffer (TLB) state must be saved and restored.
- **NUMA Penalties:** In Non-Uniform Memory Access (NUMA) architectures, memory is divided into nodes assigned to specific sockets. If a thread is moved to a different socket, accessing its originally allocated memory becomes significantly slower due to inter-socket communication links (e.g., Intel UPI, AMD Infinity Fabric).

1.2 Enforcing Affinity

To prevent migration, we can explicitly bind threads to specific hardware contexts (cores or hyper-threads). This is known as **Thread Affinity**.

In Linux environments, this is often handled using the `sched_setaffinity` system call or through library wrappers. In modern C++, we access the underlying OS handle via `.native_handle()`.

2. Memory Consistency Models

Cache coherence (discussed in the previous chapter) ensures that all cores see the same value for a *single* memory location. However, it does not guarantee the order in which memory operations across *multiple* locations become visible to different cores. This is dictated by the **Memory Consistency Model**.

2.1 Sequential Consistency (SC)

The most intuitive model, proposed by Lamport. A system is sequentially consistent if the result of any execution is the same as if the operations of all processors were executed in some sequential order, and the operations of each individual processor appear in this sequence in the order specified by its program.

- **Drawback:** Enforcing SC completely prevents the compiler and the CPU from reordering memory instructions for optimization (e.g., storing to a write buffer while a load completes).

2.2 Relaxed Memory Models

To achieve high performance, modern hardware (like x86, and particularly ARM) reorders loads and stores.

To write correct lock-free code, developers must use **Memory Barriers** (or fences) to explicitly enforce ordering where necessary.

2.3 C++11 Memory Model

C++ provides a standard way to deal with memory ordering using `std::atomic` and `std::memory_order` flags:

- `memory_order_relaxed`: No synchronization or ordering constraints, only atomicity for the specific operation.
- `memory_order_acquire` / `memory_order_release`: A store with `release` synchronizes with a load with `acquire` on the same atomic variable. All memory writes before the release become visible to the thread doing the acquire.
- `memory_order_seq_cst`: Enforces strict sequential consistency (the default, but most expensive).

3. Mitigating False Sharing

As introduced previously, false sharing occurs when independent variables accessed by different threads reside on the same cache line (typically 64 bytes).

To solve this, we must enforce spatial separation of data in memory using alignment directives.

```cpp
#include <iostream>
#include <thread>
#include <vector>
#include <atomic>
#include <pthread.h>

// align to 64 bytes to prevent false sharing between array elements
struct alignas(64) thread_state {
    std::atomic<int> counter;
};

void compute_work(int id, thread_state* state) {
    // get native handle to pin thread to specific cpu core
    pthread_t thread = pthread_self();
    cpu_set_t cpuset;
    CPU_ZERO(&cpuset);
    CPU_SET(id, &cpuset);

    // enforce affinity
    pthread_setaffinity_np(thread, sizeof(cpu_set_t), &cpuset);

    // execute workload using relaxed memory order for pure counters
    for(int i = 0; i < 100000; ++i) {
        state[id].counter.fetch_add(1, std::memory_order_relaxed);
    }
}

int main() {
    const int num_threads = 4;
    // allocated contiguous memory, but alignas guarantees 64-byte spacing
    thread_state states[num_threads];
    std::vector<std::thread> pool;

    for (int i = 0; i < num_threads; ++i) {
        states[i].counter.store(0, std::memory_order_relaxed);
        pool.emplace_back(compute_work, i, states);
    }

    for (auto& t : pool) {
        t.join();
    }

    return 0;
}
```

By manually orchestrating memory alignment and thread placement, the abstractions defined by parallel skeletons can be implemented with minimal hardware-level interference, achieving near-linear speedups bounded only by Amdahl's fraction.